Why Elimination Works

Adding equations, removing a variable, and understanding scaling

The big idea

When two equations are true at the same point, we can add their left sides and add their right sides. The new equation is also true at that point. Our goal is to combine equations so that one variable disappears.

1 Start with something familiar

$x = 3$ and $y = 5$

Add the equations: $x + y = 8$

The point (3, 5) satisfies all three equations.

2 Make a variable disappear

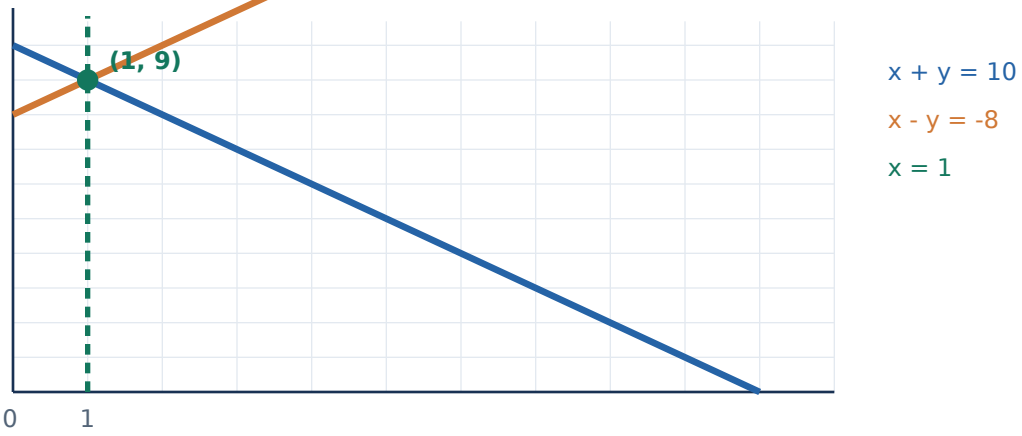
Suppose the two equations are:

$x + y = 10$

$x - y = -8$

Add: $2x + 0y = 2$ so $x = 1$

Why did y disappear? Because +y and -y are opposites, and their sum is zero. Substituting $x = 1$ into $x + y = 10$ gives $y = 9$. The solution is (1, 9).



The green vertical line $x = 1$ passes through the intersection of the original lines.

When adding does not help

3 Adding creates a true equation - but not always a simpler one

$2x + 3y = 1$

$x + y = 9$

Add: $3x + 4y = 10$

The new equation still has both x and y. It passes through the common solution, but it has not made solving easier. We need opposite coefficients before adding.

4 Why are we allowed to scale an equation?

Multiplying every term on both sides by the same nonzero number does not change which points satisfy the equation.

$x + y = 2$

Multiply both sides by 2: $2x + 2y = 4$

These are two ways of writing the same line.

x	y	x + y	2x + 2y
0	2	2	4
1	1	2	4
2	0	2	4
-1	3	2	4

5 Choose a multiplier with a purpose

Return to the equations $2x + 3y = 1$ and $x + y = 9$. We want the second equation to have $-3y$ so it cancels the $+3y$ in the first.

$2x + 3y = 1$

Multiply the entire second equation by -3:

$-3x - 3y = -27$

Add: $-x = -26$ therefore $x = 26$

Substitute: $26 + y = 9$ so $y = -17$

Check: $2(26) + 3(-17) = 1$; and $26 + (-17) = 9$. Both original equations are true.

How to decide what to scale

6 Sometimes scale one equation; sometimes both

Scale one equation when one coefficient is already a convenient multiple of the other.

$2x + 3y = 7$

$x - y = 1 \quad \rightarrow$ multiply by 3 $\rightarrow \quad 3x - 3y = 3$

Add: 5x = 10, so x = 2; then y = 1.

Scale both equations when neither coefficient is a convenient multiple of the other.

$2x + 3y = 12$

$3x - 2y = 13$

Multiply first by 2: $\quad 4x + 6y = 24$

Multiply second by 3: $\quad 9x - 6y = 39$

Add: 13x = 63, so x = 63/13. Substitute back for y = 10/13.

Teacher note: These numbers are intentionally not whole numbers. The cancellation method works just as well with fractional solutions.

7 Ask students to explain, not just calculate

Question A. Why is 4x + 4y = 8 the same line as x + y = 2?

Question B. Why does adding x + y = 10 and x - y = -8 eliminate y?

Question C. In 4x + 2y = 12 and 3x - y = 1, which equation would you scale? By what number?

Question D. Does adding two equations always eliminate a variable? Explain.

Answers for Tammy

A. Every term was multiplied by 4, preserving the same solutions. B. The y-coefficients are opposites. C. Multiply the second equation by 2, then add to eliminate y. D. No; cancellation requires opposite coefficients (or matching coefficients if subtracting).

Remember: Scaling by a nonzero number keeps the same line. Adding equations creates a new equation that every common solution must satisfy. Elimination uses both ideas to remove one unknown.